

- 12 (a)** for received signal, $28 = 10 \lg(P / \{0.36 \times 10^{-6}\})$
 $P = 2.3 \times 10^{-4} \text{ W}$ C1
A1 [2]
- (b)** loss in fibre $= 10 \lg(\{9.8 \times 10^{-3}\} / \{2.27 \times 10^{-4}\})$
 $= 16 \text{ dB}$ C1
A1 [2]
- (c)** attenuation per unit length $= 16 / 85$
 $= 0.19 \text{ dB km}^{-1}$ A1 [1]